

Experiment 2

Ayush Chandra *
22MS107

Susnata Chattopadhyay †
22MS077

Niravra Nag ‡
22MS072

*Indian Institute of Science Education and Research, Kolkata,
Mohanpur, West Bengal, 741246, India.*

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1 Aim

In this experiment we aim to study the input and output characteristics of an NPN Bipolar Junction Transistor in common emitter (CE) configuration.

2 Theory

What are BJTs?

A transistor is the solid state counterpart of vacuum triode. It is considered to be one of the most important developments in electronics with huge applications – in amplifiers, oscillators, gates etc. The *Bipolar Junction Transistors (BJT)* was invented by W.Shockley in 1951. It is an extension of Junction diode and consists of a single crystal of semiconductor with 2 p - n junctions. In this case, we have a layer of p -type material sandwiched between 2 n -type layers, thus the name NPN transistor(Fig 1).

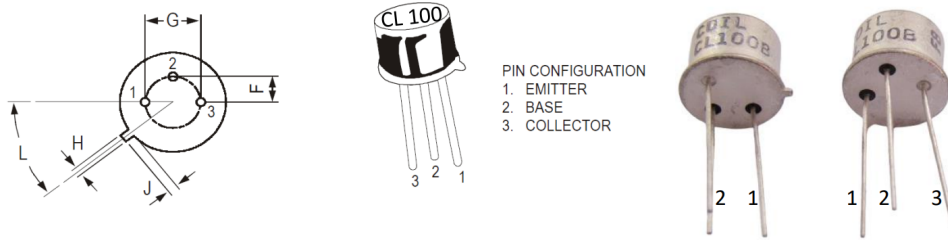


Figure 1: Configuration of the NPN transistor used in the laboratory. Source: Lab Manual

Transistor operation and Current components

In this experiment we study the characteristics curve of an NPN transistor in CE configuration. In order to understand this, it is crucial to know the operation of a transistor. We denote the *emitter-base* junction as J_E and the *collector-base* junction as J_C .

Here the majority electrons (Fig 2) from emitter are injected into the base and majority holes from base are injected into emitter region. The two constitutes the emitter current as $I_E = I_{nE} + I_{pE}$. Since the doping of emitter region is much higher than that of the base, $I_{nE} \gg I_{pE}$ and we have $I_E \approx I_{nE}$. Thus the emitter current here is almost entirely due to electrons moving from emitter to base. The

*ac22ms107@iiserkol.ac.in

†sc22ms077@iiserkol.ac.in

‡nn22ms072@iiserkol.ac.in

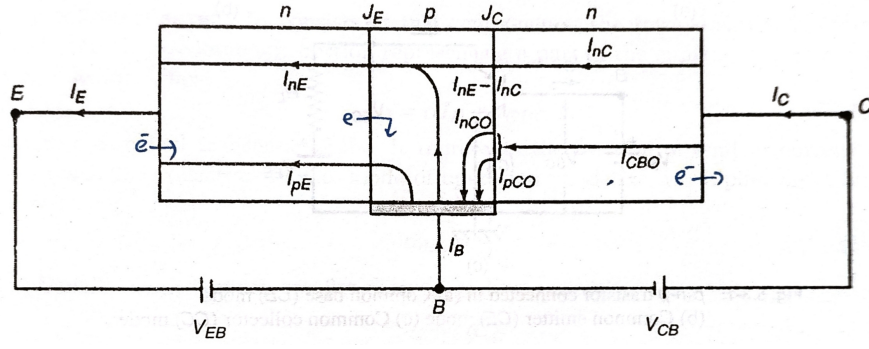


Figure 2: Current components in an npn transistor with J_E forward biased and J_C is reverse biased.

injected electrons diffuse through base towards J_E . A few of the injected electrons are also lost due to recombination with majority holes in the base.

The electrons reaching J_E are collected by the collector and give rise to the current component I_{nC} . The difference $I_{nE} - I_{nC}$ constitutes a part of base current (I_B). Since the collector junction is reverse biased there is also reverse saturation current through this junction even when the emitter is open. The collector current consists of two components, the *reverse saturation current* I_{CBO} and the predominant portion representing *a part of the emitter current* reaching the collector.

$$I_C = \alpha I_E + I_{CBO} \equiv I_{nC} + I_{CBO}$$

$$I_{CBO} = I_{nCO} + I_{pCO}$$

I_{nCO} is due to movement of *minority electrons* from base to collector, while I_{pCO} is due to movement of minority holes from collector to base. Hence we have the relations:

$$I_E = I_B + I_C$$

Input Characteristics in CE mode

The variation of the *input* base current (I_B) with input base-emitter voltage V_{BE} , with *output* collector emitted voltage V_{CE} as parameter, gives the CE **input characteristics**. For a typical curve, we expect for a constant V_{BE} , I_B to decrease with increase in V_{CE} . Increase in V_{CE} (means more electrons flowing to J_C), decreases the effective base width and hence the probability of recombination of the injected carriers in the base. Hence recombination base current is expected to decrease and so does I_B .

Output Characteristics in CE mode

The CE output characteristics are plotting output current I_C as a function of output voltage V_{CE} , taking input base current I_B as a parameter. The family of curves maybe divided into 3 regions - active, cutoff and saturation regions (Fig 3).

Active region In this region, the J_E is forward biased and the collector junction is reverse biased. In the figure (Fig 3) it is the region to the right of the ordinate $V_{CE} = V_{CE,sat}$ and above the characteristic for $I_B = 0$. $V_{CE,sat}$ is typically 0.1 to 0.2 V for low power transistors. In active region the curves are expected to be almost straight and equispaced. If a transistor is to be used as an amplifier with minimum distortion its operation should be restricted to the active region. In this region, I_C increases significantly with the increase in V_{CE} . This can be explained by as

$$I_C = \beta I_B + (1 + \beta) I_{CBO}$$

where $\beta = \frac{\alpha}{1-\alpha}$.

Due to *Early effect*¹ of collector junction is increase α increases slightly with the increase in V_{CE} . A small change in α causes very large change in β . For example, if α changes from 0.98 to 0.99, β changes

¹ If the reverse bias voltage $|V_{CB}|$ of J_E is increased, the *width of the depletion region* of collector base junction increases. This reduces the *effective width of base region* ($W'_b < W_b$). This change of base width by collector voltage is known as **Early Effect**

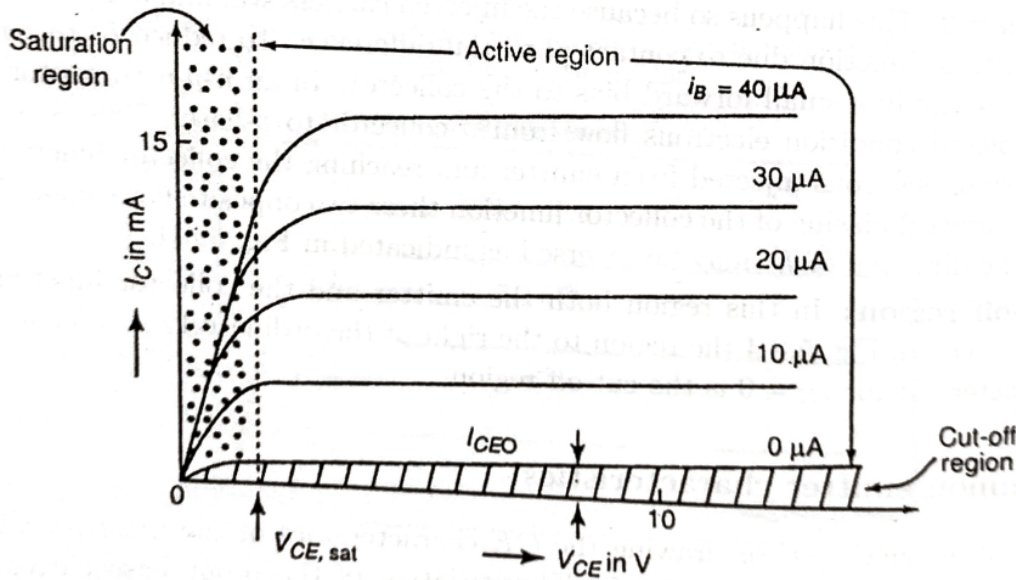


Figure 3: Typical CE output characteristics curves of an n-p-n Si transistor. Source: B. Ghosh, 2021

from 49 to 99. For this CE output curves have significant upward slope. If V_{CE} is increased very much there may be collector junction breakdown causing sharp rise in I_C . But it may damage the transistor.

Cutoff region With $I_B = 0$, $I_C = (1 + \beta)I_{CBO} = I_{CBO}$ is the collector to emitter current with the base open. Since $\beta \gg 1$, I_{CBO} has significant value and hence it is not enough to make $I_B = 0$ to cut off the transistor.

Saturation region In the saturation region both the emitter and collector junctions are forward-biased by at least the cutin voltage (V_γ). Since the forward-biasing voltages V_{EB} and V_{CB} are only a few tenths of a volt, $V_{CE,sat} = V_{CB} - V_{EB}$ is also a few tenths of a volt. So, in Fig 3, the region to the left of the ordinate $V_{CE} = V_{CE,sat}$, where all curves merge and fall rapidly towards origin is the saturation region. In this region collector current becomes almost independent of base current, for a given load and collector supply.

For a transistor operating in the saturation region the quantity $\frac{V_{CE,sat}}{I_C}$ is called *CE saturation resistance* and is designated by R_{CES} or R_{CS} . It is useful in the sense that to the left of knee the characteristic curve for a given I_B can be approximated, at least roughly, by a straight line of slope R_{CES}^{-1} . Its value is of the order of a few ohm.

3 Data Analysis

Input Characteristics

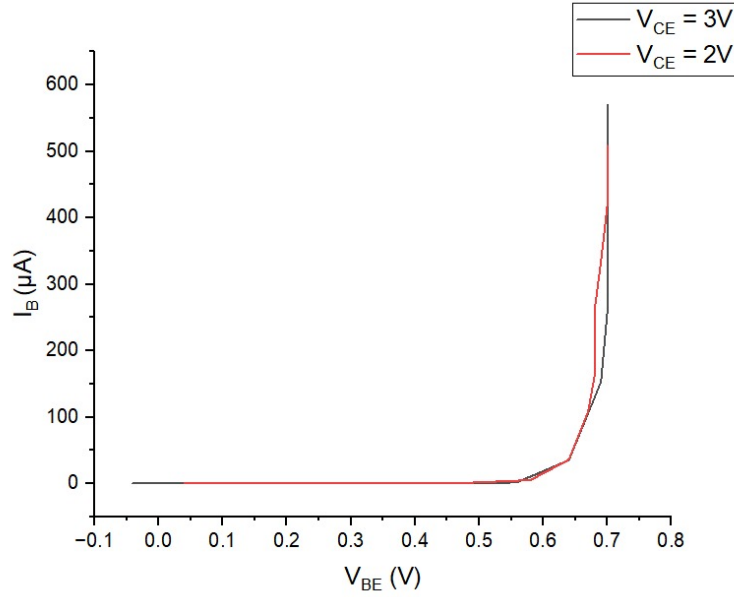


Figure 4: Input Characteristics: I_B vs. V_{BE} for different V_{CE} values.

The input characteristics depict the variation of the base current (I_B) with the base-emitter voltage (V_{BE}) at different collector-emitter voltages (V_{CE}). The data from Table 2 was used to plot the curves for V_{CE} values of 2V and 3V respectively in Fig 4. The following trends were observed:

- I_B remains negligible for small V_{BE} values.
- A significant rise in I_B occurs once V_{BE} crosses the threshold (typically around 0.6V for a silicon transistor).
- For higher V_{CE} , I_B is slightly lower due to the Early effect reducing base recombination.

Output Characteristics

The output characteristics show the relationship between the collector current (I_C) and the collector-emitter voltage (V_{CE}) for different I_B values. Data from Tables 4, 3, 5 were used to plot I_C vs V_{CE} for $I_B = 10\mu A$, $20\mu A$, and $30\mu A$ respectively in Fig 5. We observe the following:

- In the *active region*, I_C increases with I_B but remains almost independent of V_{CE} beyond a certain point.
- The *saturation region* is characterized by a rapid drop in I_C as V_{CE} decreases below a threshold (around 0.3 V).

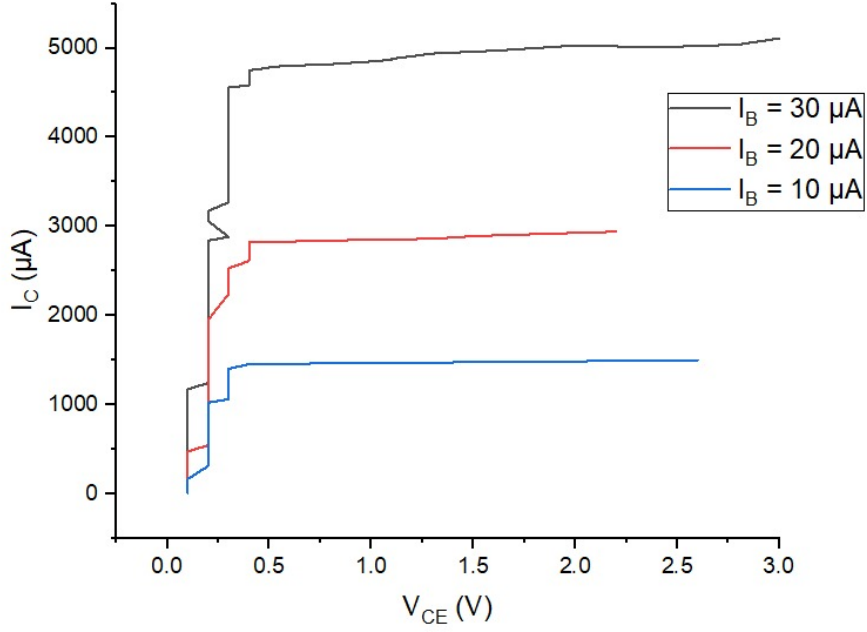


Figure 5: Output Characteristics: I_C vs. V_{CE} for different I_B values.

Calculation of Current Gain (β)

The current gain of the transistor is given by:

$$\beta = \frac{I_C}{I_B}$$

Using the data for V_{CE} around 2V, the β values were computed for different I_B values:

I_B (μA)	I_C (μA)	$\beta = \frac{I_C}{I_B}$
10	1486	148.6
20	2940	147.0
30	5030	167.6

Table 1: Current Gain (β) calculations at $V_{CE} = 5V$

Hence we calculate, $\beta = 154.4 \pm 11.46$

4 Sources of Error

- Loose connections or improper placement of wires on the breadboard could introduce additional resistance and voltage drops.
- The output of the given voltage source was not steady and was fluctuating.
- Our observation varies from theoretical expectations as our transistor is non-ideal.

5 Conclusion

The analysis of the npn BJT in the common emitter configuration confirms its expected behavior in different operating regions. The input and output characteristics validate the transistor's performance, with a predictable current gain values. The measurement of $\beta = 154.4 \pm 11.46$ further support our experimental findings.

References

- *B. Ghosh*, Fundamental Principles of Electronics 2021. ISBN: 9788187134398

Appendix

	$V_{CE} = 2V$		$V_{CE} = 3V$	
V_{BB} (V)	V_{BE} (V)	I_B (μA)	V_{BE} (V)	I_B (μA)
0.0	0.04	0	0.04	0
0.1	0.23	0	0.2	0
0.2	0.32	0	0.34	0
0.3	0.4	0	0.42	0
0.4	0.48	1	0.54	1
0.5	0.58	5	0.56	2
0.6	0.64	36	0.64	35
0.7	0.67	108	0.67	105
0.8	0.68	161	0.69	153
0.9	0.68	260	0.7	259
1.0	0.69	334	0.7	345
1.1	0.7	420	0.7	458
1.2	0.7	509	0.7	486
1.3	0.68	617	0.7	570

Table 2: Experimental Data for Input Characteristic

$I_B = 20\mu\text{A}$					
V_{CC} (V)	V_{CE} (V)	I_C (μA)	V_{CC} (V)	V_{CE} (V)	I_C (μA)
0.0	0.1	0	2.1	0.2	1756
0.1	0.1	115	2.2	0.2	1886
0.2	0.1	217	2.3	0.2	1950
0.3	0.1	313	2.4	0.3	2240
0.4	0.1	382	2.5	0.3	2340
0.5	0.1	470	2.6	0.3	2410
0.6	0.2	544	2.7	0.3	2410
0.7	0.2	642	2.8	0.3	2530
0.8	0.2	734	2.9	0.4	2610
0.9	0.2	767	3.0	0.4	2700
1.0	0.2	861	3.1	0.4	2820
1.1	0.2	921	3.2	0.4	2830
1.2	0.2	1028	3.3	0.5	2830
1.3	0.2	1090	3.4	0.6	2830
1.4	0.2	1190	3.5	0.8	2840
1.5	0.2	1275	3.8	1.0	2850
1.6	0.2	1353	4.0	1.3	2860
1.7	0.2	1451	4.3	1.5	2890
1.8	0.2	1489	4.7	1.9	2920
1.9	0.2	1603	5.0	2.2	2940
2.0	0.2	1739			

Table 3: Data for Output Characteristics for $I_B = 20\mu A$

$I_B = 10\mu\text{A}$		
V_{CC} (V)	V_{CE} (V)	I_C (μA)
0	0.1	0
0.1	0.1	133
0.2	0.1	164
0.3	0.2	308
0.4	0.2	365
0.5	0.2	472
0.6	0.2	520
0.7	0.2	573
0.8	0.2	645
0.9	0.2	711
1.0	0.2	838
1.1	0.2	916
1.2	0.2	1022
1.3	0.3	1060
1.4	0.3	1141
1.5	0.3	1245
1.6	0.3	1291
1.7	0.3	1375
1.8	0.3	1399
1.9	0.4	1452
2.0	0.6	1456
2.2	0.7	1459
2.5	1.1	1464
2.7	1.2	1469
3.0	1.5	1475
3.5	2.1	1486
4.0	2.6	1495

Table 4: Data for Output Characteristics for $I_B = 10\mu\text{A}$

$I_B = 30\mu A$											
V_{CC} (V)	V_{CE} (V)	I_C (μA)	V_{CC} (V)	V_{CE} (V)	I_C (μA)	V_{CC} (V)	V_{CE} (V)	I_C (μA)	V_{CC} (V)	V_{CE} (V)	I_C (μA)
0.0	0.1	0	2.1	0.2	1795	4.2	0.3	3880	6.7	1.8	5000
0.1	0.1	130	2.2	0.2	1931	4.3	0.3	4020	7.0	2.0	5030
0.2	0.1	227	2.3	0.2	19365	4.4	0.3	4110	7.4	2.4	5010
0.3	0.1	282	2.4	0.2	2230	4.5	0.3	4200	7.7	2.8	5040
0.4	0.1	373	2.5	0.2	2300	4.6	0.3	4240	8.0	3.0	5110
0.5	0.1	445	2.6	0.2	2420	4.7	0.3	4380	8.5	3.5	5150
0.6	0.1	444	2.7	0.2	2570	4.8	0.3	4450	9.0	3.9	5230
0.7	0.1	631	2.8	0.2	2630	4.9	0.3	4560			
0.8	0.1	699	2.9	0.2	2720	5.0	0.4	4580			
0.9	0.1	800	3.0	0.2	2840	5.1	0.4	4690			
1.0	0.1	886	3.1	0.3	2880	5.2	0.4	4750			
1.1	0.1	952	3.2	0.2	3060	5.3	0.5	4780			
1.2	0.1	1036	3.3	0.2	3090	5.4	0.5	4790			
1.3	0.1	1170	3.4	0.2	3170	5.6	0.7	4810			
1.4	0.2	1242	3.5	0.3	3270	5.7	0.8	4820			
1.5	0.2	1309	3.6	0.3	3410	5.8	1.0	4850			
1.6	0.2	1413	3.7	0.3	3500	5.9	1.1	4870			
1.7	0.2	1450	3.8	0.3	3520	6.0	1.1	4880			
1.8	0.2	1538	3.9	0.3	3610	6.1	1.2	4910			
1.9	0.2	1624	4.0	0.3	3690	6.2	1.3	4940			
2.0	0.2	1755	4.1	0.3	3830	6.5	1.6	4970			

Table 5: Data for Output Characteristics for $I_B = 30\mu A$